

WENDELL LEAO

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PROFESSIONAL SUMMARY

Senior Unity & C# Game Programmer with 5 years of professional experience working within **international** distributed teams and using **English** daily. Proven track record engineering, optimizing, and scaling high-performance **multiplayer** and **data-heavy** systems for **Mobile** and **PC (Steam)**. Recognized for **leading** architectural migrations, eliminating critical performance bottlenecks, and doubling runtime performance (**FPS**) in **AA** titles. Expert in multiplayer state synchronization, advanced UI architectures (**UI Toolkit/uGUI**), and designing backend-driven pipelines that handle **thousands** of dynamic entities seamlessly.

SKILLS

- **Languages & Core Tech:** C#, Engine Architecture, Gameplay Systems, Scriptable Objects, Data-Oriented Design.
- **Architecture & Patterns:** SOLID Principles, Dependency Injection, State Machines, Factory, Command, Observer.
- **Multiplayer & Networking:** Mirror, Photon Fusion/Quantum, Netcode for GameObjects, State Synchronization, RPC Handling, Latency Mitigation, UGS (Lobby/Auth).
- **UI & Frontend Engineering:** UI Toolkit (Advanced Diegetic Interfaces), uGUI, IMGUI, Data-Driven UI Architecture.
- **Performance & Optimization:** CPU/GPU Profiling (Unity Profiler, Frame Debugger), Memory Management, Addressables, Asset Bundles, UniTask (Asynchronous Programming).
- **Graphics & Pipelines:** Scriptable Render Pipeline (URP/HDRP Migration & Optimization), Shader Graph, Custom Materials, Cinemachine.
- **Tools & Integrations:** Steamworks API, RESTful APIs (Swagger), Git/Git-Flow, CI/CD Pipelines, Custom Editor Tooling.

WORK EXPERIENCE

Senior Unity Developer (Contractor) - BGNS Studios, Serbia

May 2024 - Present

[Draft Fever Bowl](#) - PC (AA)

- **Frontend Technical Leadership:** Acted as the primary **frontend technical representative** for an **AA PC** title, establishing architectural standards, conducting peer **code reviews**, and **leading** cross-department synchronization between frontend, backend, and simulation teams.
- **Data System Optimization:** Designed and engineered a data-driven drafting and scouting system using **Virtualized Lists** to handle over **2,000** dynamic entities in real-time; **doubled** performance from **40 FPS** to **80-100 FPS** on critical screens.
- **Render Pipeline Migration (HDRP to URP):** Directed a full **Scriptable Render Pipeline** migration, refactoring rendering-dependent systems and custom shaders; boosted menu performance to **80 FPS** and stabilized simulation performance from **20-30 FPS** to a solid **60 FPS**.
- **Runtime Asset Pipelines:** Redesigned the runtime character assembly architecture using dynamic **Skinned Mesh Renderer** swapping, reducing initial game loading times by over **50%** (from **1 minute** down to **30-25s**) and eliminating pre-loaded memory variants.
- **Asynchronous System Engineering:** Optimized a complex emblem and stadium customization system by implementing asynchronous processing via **UniTask** and semaphore synchronization, completely eliminating **frame drops** during live texture and camera capture generation.

[PIX Football Manager](#) - PC

- **Closed Beta Support & Live Hotfixes:** Supported the game's **closed beta launch** by closely monitoring player experiences and rapidly deploying critical **hotfixes** based on live community **feedback** to ensure stability and gameplay polish.
- **Complex UI Architecture:** Built highly interactive **diegetic** interfaces using **UI Toolkit** for a **PC** football manager title (**3.5M+** views on **YouTube**), architecting new layout flows and executing comprehensive revamps of legacy screens to deliver a significantly more intuitive and fluid user experience.
- **Backend Integration & State Management:** Utilized **Swagger UI** to map and implement **RESTful API** endpoints (**GET, POST, PUT, DELETE**) within **Unity**, synchronizing player account data, balance sheets, and complex contract life cycles (hiring, firing, and expiration/renewal mechanics).

Amazon Warriors - Mobile

- **Mobile Performance & Memory Management:** Authored a comprehensive, data-driven **UI framework** used across multiple mobile titles; implemented **Object Pooling** for **procedural** terrain and props, eliminating execution spikes and stabilizing mobile gameplay at a locked **30 FPS**.
- **Procedural Systems Framework:** Restructured legacy codebases under tight **LiveOps** deadlines to implement procedural level generation, biome transitions, and dynamic obstacle systems.
- **LiveOps & Backend Integration:** Integrated **monetization** flows, **advertisements**, and external **telemetry** services via **Unity Gaming Services (UGS)**, leveraging **Remote Config** and **Analytics** to drive player engagement.

Rogue Masters - PC (AA)

- **Multiplayer Infrastructure Maintenance:** Took ownership of a mature multiplayer architecture using **Mirror (P2P)** for an **AA Steam** title supporting **6** players/session, successfully expanding **client-server** communication, **RPC** handling, and custom **matchmaking/lobby** filtering systems.
- **Network Assembly Optimization:** Refactored critical multiplayer synchronization codebases, redesigning the character assembly pipeline to use lightweight ID-driven updates instead of heavy synchronization, completely **eliminating frame drops** during real-time equipment changes for multiple players.
- **Startup Architecture Refactor:** Resolved legacy engine-freezing bugs by implementing an asynchronous **bootstrap/startup** scene, organizing the initialization sequence for **Steamworks**, **SDKs**, and **backend** services to dramatically increase game stability.
- **State Synchronization & Bug Diagnosis:** Specialized in multiplayer gameplay replication (projectiles, **AI** behaviors, and physics trajectories), utilizing the **Unity Profiler** in advanced mode to diagnose and eliminate deep legacy exceptions and multiplayer desynchronization bugs.

UniqKiller - Mobile

- **Advanced Networking & Gameplay (Photon Fusion):** Expanded the weapon system for dedicated **12-player** servers by developing and synchronizing multiplayer-ready weapons, including implementing projectile trajectories for a grenade and integrating asset/mechanic pipelines for a sniper rifle.
- **AI & Pathfinding Optimization:** Configured and baked terrain **NavMesh** for competitive maps, ensuring seamless bot navigation and preventing stuck behaviors while optimizing performance to eliminate redundant **pathfinding** calculations.

Pet Shop Fever - Mobile

- **Mobile Memory Optimization:** Scaled memory efficiency on a pet shop manager title (**1M+** downloads on **iOS/Android**) by architecting an **additive-scene** UI framework, streaming interfaces dynamically to minimize memory footprint on low-end devices.
- **Asset Pipeline Restructuring:** Achieved significant build size reductions and optimized loading screen times by transitioning the project's asset management workflow to **Unity Addressables** and **Asset Bundles**.

Games By MaxPro - Mobile

- **Gameplay & Hardware Integration:** Worked on an existing custom hardware integration pipeline for a fitness-focused title, building gameplay mechanics, menu flows, and UI systems that **transformed external physical** activity data streams into active in-game interactions.

LANGUAGES

Portuguese: Native | English: Professional proficiency (advanced) | Spanish: Beginner

EDUCATION

Bachelor of Technology in Digital Game Development - FMU, Brazil (Feb 2019 - July 2021)